Calibrated determinant guards for Diophantine ZFC consistency

A quartic in 38 integers, a sextic in 32, an octic in 27, and two positive-variable improvements

Prepared for Vladimir Reshetnikov and the ProveIt repository

AI-assisted mathematical research; exact finite certificates included

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Abstract

The principal result is a new way to remove positivity witnesses without raising the degree of a Diophantine equation. The canonical native encoding already fixes the leading digits of two packed integers. Fixed rescalings place those canonical witnesses in a narrow region where one quadratic determinant is positive. Positivity of that determinant, for arbitrary signed assignments, recovers both common-sign conditions previously imposed separately. Replacing two ternary positivity representations by one reduces the inspected 41-integer quartic to **38 unrestricted integer variables** and 25 quadratic residuals. Exact coordinate eliminations give a sextic in **32 integers** and an octic in **27 integers**.

Two independent companions give twelve positive variables at degree **110** in **102** binary operations, and nine positive variables at degree **1,379,730** in **478** operations. Every coefficient is fixed by supplied finite data. The new proofs compose with explicitly named native-compiler, Pell and syntactic-refutation results in ProveIt to give the requested ZFC-provable unsolvability/consistency equivalences. No consistency assumption, global optimum, literature-priority claim, or new end-to-end proof-assistant certificate is asserted.

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1 What is constructed and what improves

The target is an ordinary polynomial with fixed integer coefficients, not an exponential equation and not a formula with hidden quantifiers. The convention for consistency is syntactic:

$$\mathrm{Con}(\mathsf{ZFC}) \;\equiv\; \neg\exists p\ \mathrm{Ref}_{\mathsf{ZFC}}(p),$$

where $\mathrm{Ref}_{\mathsf{ZFC}}$ is the fixed recursive refutation predicate used by the repository's calibrated endpoint. The desired theorem is an equivalence *provable in ZFC*; it does not ask ZFC to prove either consistency or the absence of zeros separately.

The immutable comparison snapshot is `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`. In particular, the comparison includes all eight wave-9 packages 60–67. The collection README's older frontier is insufficient for this comparison. The main previous integer points are package 64's 41-variable quartic and package 62's 35-variable sextic and 30-variable octic [2, 3]. Table 1 records the changes, with different witness domains kept explicit.

Domain	Unknowns	Degree	Operations	Prior point	Definition
$\mathbb{Z}$	**38**	4	–	41 / 4	Q_{38}
$\mathbb{Z}$	**32**	6	–	35 / 6	S_{32}
$\mathbb{Z}$	**27**	8	–	30 / 8	O_{27}
$\mathbb{Z}_{>0}$	12	**110**	**102**	118 / 104 ops	P_{12}
$\mathbb{Z}_{>0}$	9	**1,379,730**	**478**	1,494,006 / 487 ops	F_9
$\mathbb{N}$	9	1,379,730	485	separate translation	$F_{9,\mathbb{N}}$

Table 1: New explicit equations against the inspected wave-9 frontier. The first three prior points are unknowns / degree. Operation counts for the integer forms have not been optimized.

The variable reductions are $3/41 \approx 7.32\%$, $3/35 \approx 8.57\%$, and $3/30 = 10\%$, respectively. The nine-positive companion reduces the best inspected degree by 114,276, or approximately 7.65%, and saves nine operations relative to that same predecessor. Package 62's 453-operation nine-positive equation still has a lower operation count at higher degree. The ten-unrestricted-variable frontier is not improved.

There is no agreed scalar measure of "simplest." In this report a degree reduction, a variable reduction, and a shorter evaluator are separate achievements. Binary arithmetic counts charge every $+, -, \times$, including squarings and multiplication by fixed numerals; reuse and constant leaves are free. They are not bit-complexity estimates. All quoted degrees count only existential unknowns, after specializing the fixed coefficient ports.

Theorem 1.1 (Constructed consistency equations)**.** *With the complete coefficient bindings in the archive, the constructions below satisfy, by composition with the named inherited compiler and calibration theorems,*

$$\begin{aligned}
\mathsf{ZFC} &\vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^{38}\ Q_{38}(\mathbf{x}) = 0) \leftrightarrow \mathrm{Con}(\mathsf{ZFC}),\\
\mathsf{ZFC} &\vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^{32}\ S_{32}(\mathbf{x}) = 0) \leftrightarrow \mathrm{Con}(\mathsf{ZFC}),\\
\mathsf{ZFC} &\vdash (\neg\exists \mathbf{x} \in \mathbb{Z}^{27}\ O_{27}(\mathbf{x}) = 0) \leftrightarrow \mathrm{Con}(\mathsf{ZFC}).
\end{aligned}$$

The analogous assertions hold for P_{12} over $\mathbb{Z}_{>0}^{12}$, F_9 over $\mathbb{Z}_{>0}^{9}$, and its explicitly shifted companion over $\mathbb{N}^9$.

The proof dependency is substantive: this article gives complete ordinary proofs of its new reductions, and imports the established native arithmetic and refutation calibration used by the cited packages. It does not replace those dependencies by numerical tests. Section 8 separates that mathematical composition from the repository's still distinct task of exporting a checked derivation in its object-language ZFC calculus.

2 The new principle: one determinant for two sign conditions

For integer variables define

$$\mathrm{Pos}(x,y,z) = 1 + x^2 + y^2 + z(z+1).$$

It is always positive, since $z(z+1) \geq 0$ for every integer z. Conversely every positive integer n is a value of Pos: Legendre's three-square theorem applies to $4n - 3 \equiv 1 \pmod 4$, which is not of the excluded form $4^a(8b+7)$. Exactly one of the three squares is odd. Writing that term as $(2z+1)^2$ and the other two as $(2x)^2, (2y)^2$ gives the claimed representation.

The previous quartic used two independent triples to impose

$$F\alpha = \mathrm{Pos}(f_1,f_2,f_3), \qquad FC = \mathrm{Pos}(c_1,c_2,c_3), \qquad q = F + \alpha + C. \tag{1}$$

These rows require six private integer witnesses. The new idea is to use the special location of *canonical* accepting witnesses to choose a stronger quadratic condition that requires only one triple.

Lemma 2.1 (Weighted determinant guard)**.** *Let A_0, B_0 be fixed integers satisfying $A_0, B_0 \geq 2$. For integers q, F, C put*

$$U = A_0F, \quad V = B_0C, \quad W = q - U - V, \qquad \mathcal{G} = UV - W^2.$$

If $\mathcal{G} > 0$, then $q, F, C, q - F - C$ are nonzero and have the same sign.

Proof. $UV > W^2 \geq 0$ implies that U, V have one common nonzero sign. Set $L = |U| + |V|$. The elementary inequality $2\sqrt{|U||V|} \leq L$ gives $|W| < L/2$. Thus $q = U + V + W$ has the same sign as U, V, and

$$|q| > L/2 = (A_0|F| + B_0|C|)/2 \geq |F| + |C|.$$

So q, F, C have one sign and $|q| > |F| + |C|$, which gives the same sign to $q - F - C$. The proof does not assume $q > 0$. □

Consequently the single quadratic row

$$\boxed{A_0B_0FC - (q - A_0F - B_0C)^2 = \mathrm{Pos}(t_1,t_2,t_3)} \tag{2}$$

implies both positivity conditions in (1). One may view the left side as the determinant of $\left(\begin{smallmatrix} U & W \\ W & V \end{smallmatrix}\right)$. The linear relation to q recovers the common sign of this definite matrix.

Soundness alone is easy; completeness is the actual research issue. Arbitrarily strengthening a positivity condition can destroy every witness. Here the fixed native coefficient geometry supplies a uniform region near $U = V = q/2$, in which the determinant remains positive.

Lemma 2.2 (Uniform canonical region)**.** *Let $s > 0$, $q = 2s$, and $s \leq U < 8s/5$, $s \leq V < 6s/5$. Then*

$$UV - (q - U - V)^2 > \frac{12}{25}s^2 > 0.$$

Proof. Write $U = s + x, V = s + y$, with $0 \le x < 3s/5$ and $0 \le y < s/5$. Then $W = -x - y$ and

$$UV - W^2 = s^2 + s(x + y) - x^2 - xy - y^2 > s^2 - (9 + 3 + 1)s^2/25.$$

□

The reduction is therefore asymmetric in the useful way. An arbitrary new zero restores an old zero by Lemma 2.1; a refutation supplies fresh canonical witnesses that satisfy Lemma 2.2. Equality of the two zero sets on the same old coordinates is neither asserted nor needed.

3 Why the actual native encoding lies in that region

This section fixes the coefficients, rather than treating the guard as a family with unbound parameters. Use the *whole* source-34 tuple `mixed-parity-unrounded-1064921` [1]. Write

$$\rho = 2^{121}, \quad L_0 = 8801, \quad B = \rho^{L_0} = 2^{1064921}, \qquad \beta = B - 1.$$

The first native tile, with ordered labels $(2, 435, 3, 1)$, has clause coefficient

$$c_0 = 10553438982482419464738229862 5, \qquad 2^{96} \le c_0 < c_0 + 1 \le 3 \cdot 2^{95} < 2^{97} < \rho. \tag{3}$$

The two new fixed numerals are simply powers of two:

$$\boxed{A_0 = 2^{1113}, \qquad B_0 = 2^{666104}.} \tag{4}$$

In particular $A_0, B_0 \ge 2$. These are additional coefficients, not existential variables. The archive reconstructs all six ports and checks the four inherited raw values against their pinned SHA-256 hashes.

3.1 Leading digits of the canonical C word

A finite accepting computation supplies a padded cyclic native word with N occupied cells and $q = B^N$. At each cell j, the source construction puts one genuine tile, its occupied edge copies, and its four occupancy anchors. The largest support exponent of its native integer C_j is 3296, with coefficient one. The next possible exponent is 2195. Therefore

$$\rho^{3296} \le C_j < \rho^{3296} + \rho^{2196} \le (1 + 1/\rho)\rho^{3296}. \tag{5}$$

The rank-tuning dummy is at exponent 1063; changing its permitted value cannot affect either leading-support fact. The inequalities hold for every size of a canonical accepting rectangle, including after all dummy changes.

3.2 Leading digits of the canonical F word

Let D_C, D_R be the fixed native convolution coefficients, so $K_0 = D_C + BD_R$ and $D_R = \rho$. The rank controller gives a whole-cell temporal shift w_c with

$$R \equiv 1064921 w_c \pmod{1064921N}.$$

Thus the canonical transport word is the residue

$$F \equiv (D_C + BD_R + B^{w_c})C \pmod{q - 1}.$$

The actual block at cell j, with cyclic indices, is

$$F_j = D_C C_j + D_R C_{j-1} + C_{j-w_c}. \tag{6}$$

The largest exponent in D_C is 5495, with coefficient c_0. The unique pair reaching exponent $8791 = 5495 + 3296$ in $D_C C_j$ has coefficients c_0 and one. The other two summands in (6) have exponents at most 3297 and 3296.

It is necessary to exclude a carry into that leading coefficient. Let $S = \sum_e c_e$ be the sum of the source clause coefficients. The sum of coefficients of D_C is $4 + 2S$. Each canonical C_j coefficient is at most three; the possible larger dummy value is still only three. Hence each convolution coefficient in (6) is at most

$$3(4 + 2S) + 6 = 6S + 18 < 991(2S + 6) = \mathcal{M} < \rho.$$

The last inequality is among the reconstructed source's strict finite carry bounds; the stronger $4\mathcal{M} + 8 < \rho$ is checked. Thus no carry changes the top digit, and

$$c_0 \rho^{8791} \le F_j < (c_0 + 1)\rho^{8791}. \tag{7}$$

Every block is strictly below $B - 1$ since $8791 < 8801$ and $c_0 < \rho$. Their base-B word is therefore the canonical representative in $(0, q - 1)$; the residue calculation has not confused zero with $q - 1$.

3.3 One estimate for every accepting witness length

Put $J_N = (q - 1)/(B - 1) = \sum_{j<N} B^j$. The exponents were chosen because

$$121 \cdot 8791 + 1113 = 1064921 - 97, \qquad 121 \cdot 3296 + 666104 = 1064921 - 1.$$

Summing (5) and (7) therefore gives

$$\frac{c_0}{2^{97}} B J_N \le A_0 F < \frac{c_0 + 1}{2^{97}} B J_N, \qquad \frac{1}{2} B J_N \le B_0 C < \frac{1}{2}(1 + 1/\rho) B J_N.$$

For all $N \ge 1$, $q \le B J_N < qB/(B - 1)$. Using (3), $\rho \ge 8$ and $B \ge 16$ gives

$$\frac{q}{2} \le A_0 F < \frac{3}{4}\frac{16}{15} q = \frac{4}{5} q, \qquad \frac{q}{2} \le B_0 C < \frac{1}{2}\frac{9}{8}\frac{16}{15} q = \frac{3}{5} q.$$

Set $s = q/2$. Lemma 2.2 applies. This proves completeness of the new determinant guard for the actual coefficient tuple. The large sizes of A_0, B_0 express a known disparity of native digit positions; they do not encode the answer to the consistency question.

4 The complete 38-integer quartic

All variables in this section range over *unrestricted integers*. There are 32 arithmetic variables

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta, u$$

and six guard variables $t_1, t_2, t_3, e_1, e_2, e_3$, totaling 38. In the machine-readable packets, v, w, B_a are named `v_scale`, `w_scale`, `Baux`. For readability define the polynomial abbreviations

$$q = \beta J + 1, \qquad C = q - F - \alpha, \qquad k = \eta + \zeta, \qquad H = 4a + 3.$$

$$\mathcal{G} = A_0B_0FC - (q - A_0F - B_0C)^2,$$
$$\mathcal{R} = q(N - q - L) + F(q + L - N) + \alpha L + (M_C + qM_F)J.$$

$q, C, k, H, \mathcal{R}, \mathcal{G}$ are not additional unknowns. Every displayed residual in the following list has degree at most two:

$$r_1 = L - q^2 + 1, \qquad r_2 = N - q(L + 1),$$
$$r_3 = X - 4wq, \qquad r_4 = a - E - (2v + 1)N,$$
$$r_5 = E(X + 1) - Xa, \qquad r_6 = c - \eta - (a - E)k,$$
$$r_7 = P - E(c - \eta) - k, \qquad r_8 = \tau^2 - P^2 + Pk - 1,$$
$$r_9 = D - X - ac - \gamma H, \qquad r_{10} = \Delta - a^2 - 4a - 3,$$
$$r_{11} = C_2 - c^2, \qquad r_{12} = D^2 - \Delta C_2 - 1,$$
$$r_{13} = k - 1 - hE - \mathcal{R}, \qquad r_{14} = (K_0 + 4w)C - F - z(q - 1),$$
$$r_{15} = F_2 - f^2, \qquad r_{16} = V + c - fo,$$
$$r_{17} = V - jc + k - 1 - hE, \qquad r_{18} = V^2 + g - y^2,$$
$$r_{19} = I - (2r + 1)C_2, \qquad r_{20} = B_a - I^2,$$
$$r_{21} = B_a - \Delta(F_2 - 1), \qquad r_{22} = y^2 - B_a g - 1,$$
$$r_{23} = X - 2u^2, \qquad r_{24} = \mathcal{G} - \mathrm{Pos}(t_1, t_2, t_3),$$
$$r_{25} = \eta\zeta - \mathrm{Pos}(e_1, e_2, e_3).$$

The promised single equation is

$$\boxed{Q_{38} = \sum_{\nu=1}^{25} r_\nu^2 = 0.} \tag{8}$$

The use of a sum is just a compact specification of one polynomial. All 25 residuals and its expanded symbolic coefficient list are included. The coefficient of τ^4 is exactly one, so its degree is exactly four on the actual fixed coefficient slice.

4.1 Existence equivalence

At an integer zero of Q_{38} every residual is zero. The new guard gives $F\alpha > 0$ and $FC > 0$ by Lemma 2.1. Represent these two positive integers by the old six private Pos variables. All other rows are literally the corresponding package-64 quartic rows. Thus an integer zero of Q_{38} supplies a zero of that package's Q_{41} and, by its native/Pell soundness and K215 calibration, a finite ZFC refutation.

Conversely a refutation supplies the canonical witnesses in package 64's completeness proof. They use the whole source-34 tuple, and satisfy all retained rows, including $X = 2u^2$ and the divisible-by-four scale. Section 3 proves $\mathcal{G} > 0$ for those very witnesses, after rank tuning. A ternary Pos representation supplies the new three variables. Hence Q_{38} has an integer zero.

The quantifier order matters: one first obtains an accepting computation, then a canonical native witness, then the new Pos representation. This is why restricting the old numerical witnesses is legitimate. No search for the absence of a witness, and no assumption of Con(ZFC), occurs.

4.2 Coefficient and proof costs

The four old native numerals occupy 532,479 bytes in minimal unsigned little-endian form. The two new leaves add 83,404 bytes, for a total of 615,883 bytes. The archive also provides short

power-of-two recipes and compressed payloads. This is a real coefficient-storage tradeoff for three fewer variables at unchanged degree. It is not a claim that the expanded polynomial's total bit length decreases. In fact the J^2 coefficient of the new residual is $-\beta^2(B_0-1)^2$, so squaring that residual contributes $\beta^4(B_0-1)^4$ to the J^4 coefficient. Including the retained rows, the exact coefficient is

$$[J^4]Q_{38} = \beta^4(B_0-1)^4 + \beta^4 + \beta^2(\beta - M_F)^2.$$

It has exactly 6,924,100 bits for the supplied tuple, as checked both by exact inequalities and by reconstruction. This is a lower bound on the maximum coefficient bit length, not a claim that this coefficient is the largest. Variable count, coefficient height and arithmetic description length must therefore be assessed separately.

The inherited signed Pell argument remains unchanged. In particular the new proof does not assume a positive radix before the existing argument recovers its sign. It also preserves the corrected transport convention $(K_0+4w)C = F + z(q-1)$; a different earlier convention uses $z+1$.

5 Exact sextic and octic reductions

These reductions are purely algebraic. Whenever a residual is $x - f(\text{other variables})$, deleting it and substituting f everywhere gives a bijection of integer zero sets, with the explicit polynomial inverse restoring x. No divisibility or sign premise is needed. The verifier checks the identity at each step and also substitutes the complete final inverse into every original quartic row.

5.1 A sextic in 32 integers

First make the integral triangular change

$$\tau_{\text{old}} = \tau_{\text{tail}} + P - k.$$

Then eliminate $P = k + E(c-\eta)$. The first Pell residual becomes

$$\tau_{\text{tail}}^2 + (2\tau_{\text{tail}} - k)E(c-\eta) - 1,$$

which has degree three: its degree-four terms have cancelled exactly. Eliminate, in addition,

$$F_2 = f^2, \quad g = y^2 - V^2, \quad X = 2u^2, \quad N = q(L+1), \quad C_2 = c^2.$$

This removes six variables and six defining equations. The resulting 19 residuals have degree at most three in 32 integers. Let S_{32} be their sum of squares. The full inverse includes

$$\tau_{\text{old}} = \tau_{\text{tail}} + E(c-\eta).$$

Only the retained strong-definition row $I - (2r+1)c^2$ contains r. Its square gives

$$[r^2c^4]S_{32} = 4,$$

so $\deg S_{32} = 6$ for the authentic fixed tuple. The full literal list, expanded polynomial and inverse map are in `integer_forms/sextic32.json` and its companion files.

5.2 An octic in 27 integers, written out completely

This version removes 11 registers. It is short enough to state without any implicit source reference. Its 21 arithmetic unknowns are

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, y, \gamma, o, j, E, c, a, V, I, u,$$

and its six guard unknowns are again $t_1, t_2, t_3, e_1, e_2, e_3$. Define only polynomial abbreviations

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, \qquad k = 1 - V + jc + hE, \\ H &= 4a + 3, & \Delta &= a^2 + H, \qquad B_1 = E(c - \eta), \\ R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\ \mathcal{G} &= A_0 B_0 FC - (q - A_0 F - B_0 C)^2. \end{aligned}$$

Here B_1 is a polynomial abbreviation and is unrelated to the fixed native radix B. The 14 residuals are

$$\begin{aligned} s_1 &= 2u^2 - 4wq, \\ s_2 &= a - E - (2v+1)q^3, \\ s_3 &= E(2u^2+1) - 2u^2 a, \\ s_4 &= c - \eta - (a - E)k, \\ s_5 &= \tau^2 - (k + B_1)B_1 - 1, \\ s_6 &= (2u^2 + ac + \gamma H)^2 - \Delta c^2 - 1, \\ s_7 &= -V + jc - R, \\ s_8 &= (K_0 + 4w)C - F - z(q-1), \\ s_9 &= V + c - fo, \\ s_{10} &= I - (2r+1)c^2, \\ s_{11} &= I^2 - \Delta(f^2 - 1), \\ s_{12} &= y^2 - I^2(y^2 - V^2) - 1, \\ s_{13} &= \mathcal{G} - \mathrm{Pos}(t_1, t_2, t_3), \\ s_{14} &= \eta(k - \eta) - \mathrm{Pos}(e_1, e_2, e_3). \end{aligned} \tag{9}$$

The explicit equation is

$$\boxed{O_{27} = s_1^2 + \cdots + s_{14}^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^{27}.} \tag{10}$$

This is one convenient answer if a low degree and unrestricted integer variables are preferred over the nine-positive high-degree construction.

Restore the eliminated quartic coordinates by

$$\begin{aligned} L &= q^2 - 1, & N &= q^3, & X &= 2u^2, \\ C_2 &= c^2, & F_2 &= f^2, & g &= y^2 - V^2, \\ B_a &= I^2, & D &= 2u^2 + ac + \gamma H, & \Delta &= a^2 + 4a + 3, \\ \zeta &= k - \eta, & P &= k + E(c - \eta). \end{aligned}$$

These are integer polynomials. Substitution makes each original row identically zero or one of the 14 listed residuals, with a possible harmless sign change. Conversely projection recovers the octic coordinates. Every residual has degree at most four, and

$$[a^4 f^4] O_{27} = 1.$$

Only the strong row $I^2 - \Delta(f^2 - 1)$ can supply that monomial. Thus $\deg O_{27} = 8$, not just an upper bound. Both eliminated forms inherit the existence equivalence of Q_{38} by exact inverse maps.

6 Twelve positive variables: an adaptive carrier and a centered norm

This companion improves package 61's twelve-positive-variable equation from degree 118 and 104 binary operations to degree 110 and 102 operations [4]. The first carrier is retained; the adaptive main carrier of package 65 lowers one residual degree [5], while a centered auxiliary norm eliminates two positive witnesses with a simpler completeness argument. The witness domain remains strictly positive integers.

The coefficients in this section are the *whole compact package-44 native tuple*, independently of the source-34 tuple used by the integer constructions above:

$$B = 2^{590069}, \qquad \beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_F^{\mathrm{native}} + B - 1.$$

The last three numerals, their finite generator and their exact byte bindings are supplied in `positive12/native/`; M_F is already shifted. All four ports must be specialized together. The inherited compact adapter supplies the native computation interpretation and canonical positive witnesses; the new steps below are elementary existence reductions on that interface.

Theorem 6.1. *The polynomial P_{12} defined below has twelve strictly positive unknowns, exact total degree 110, and a literal evaluator with 102 live binary arithmetic operations. With its supplied compact coefficient binding, composition with the inherited compiler and syntactic-refutation calibration proves*

$$\mathsf{ZFC} \vdash (\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\boldsymbol{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

This is an ordinary mathematical proof using the named inherited interface, not a newly exported end-to-end proof-assistant certificate.

6.1 The complete polynomial

Supply precisely

$$j, F, \alpha, i, T, X, Y, r, \eta, \zeta, n, \gamma \in \mathbb{Z}_{>0}.$$

Every subsequent register is an integer polynomial abbreviation, not an additional existential unknown. Define

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & A &= a + 2, & H &= 4a + 3,\\
\Delta &= a^2 + H = A^2 - 1, & \ell &= EkY, & b_0 &= X + ac,
\end{aligned}$$

$$R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J, \qquad v = k - R - 1.$$

The two carriers are

$$\begin{aligned} W &= (q-1)k+1, & M_0 &= q^3(q-1), \\ B_t &= (qK_0+X)C - qF, & & \\ Z &= M_0(\ell+k+r) + (q-1)Y - q^3 W B_t, & & \\ A_0 &= Z^2 - M_0^2\big(\ell(\ell+k)+1\big), & & \qquad (11) \\ S &= 4qY, & M &= qS, \\ D_* &= Mb_0 + H(M\gamma - v + SX), & & \\ A_1 &= D_*^2 - M^2(1+\Delta c^2). & & \qquad (12) \end{aligned}$$

For the centered norm put

$$\begin{aligned} I &= ic^2, & D &= 1+\Delta I^2, & Q &= \Delta^2 I^2, & d &= Q-1, \\ b &= cT - R, & V &= Db - c, & t &= AI + n, & & \qquad (13) \\ U &= d\big((V+t)^2 + D\big) - QV^2 + 1, & & & & & & \\ L &= 2d(V+t), & N &= U^2 - L^2 D. & & & & \qquad (14) \end{aligned}$$

The requested single equation is

$$\boxed{P_{12} = (1+A_0)(1+A_1)(1+N) - 1 = 0.} \qquad (15)$$

There are no variable powers, divisions or radical operations in this specification. The literal graph is `positive12/adaptive_centered12.json`.

Since β and J are odd, q is even. Thus M_0 is even, M is divisible by four, and L is even. Consequently

$$A_0 \equiv Z^2, \qquad A_1 \equiv D_*^2, \qquad N \equiv U^2 \pmod 4.$$

If the product in (15) is one, each integer factor is 1 or -1. The latter would make its residual -2, which is not a square residue modulo four. Hence, on the stated domain,

$$P_{12} = 0 \quad \Longleftrightarrow \quad A_0 = A_1 = N = 0. \qquad (16)$$

No nonnegativity assumption on the three residuals is needed.

6.2 Why the centered norm restores positive witnesses

Unconditionally, $c \geq 3$, $I \geq 9$, $D > c+1$ and

$$t^2 > (AI)^2 > D, \qquad (AI)^2 - D = I^2 - 1.$$

At a zero of N, its denominator register L cannot vanish. Otherwise $V + t = 0$ and

$$U = dD - Qt^2 + 1 < dD - QD + 1 = 1 - D < 0,$$

contrary to $N = U^2 = 0$. Therefore $f = U/L$ is rational with integer square D, and hence is an integer. Set $y = V + t - f$. Expanding its square gives

$$QV^2 - dy^2 = 1. \qquad (17)$$

Both roots may have either sign. Nevertheless $|V| > 1$: the integer $b \le 0$ gives $V \le -c$, while $b \ge 1$ gives $V \ge D - c > 1$. Equation (17) consequently implies $d(y^2 - V^2) = V^2 - 1 > 0$. Thus

$$f_0 = |f| > 0, \qquad u = |y| - V > 0$$

restore exactly the normalized strong and auxiliary source equations:

$$f_0^2 - \Delta(ic^2)^2 = 1,$$
$$(V+u)^2 - \Delta(f_0^2 - 1)u(2V+u) = 1, \qquad V = f_0^2(cT - R) - c.$$

This handles every signed root branch. It proves an existence reduction, without asserting a bijection on arbitrary predecessor witnesses.

For completeness, the canonical source allows fresh auxiliary witnesses

$$V = \frac{\chi_{\Delta I}(R)}{\Delta I}, \qquad y = \psi_{\Delta I}(R), \qquad R \equiv 3 \pmod 4,$$

where $\chi_s(m) + \psi_s(m)\sqrt{s^2-1} = (s + \sqrt{s^2-1})^m$. The retained odd-index identities give $V \equiv -c \pmod{f^2}$ and $V \equiv -R \pmod c$. Since $\gcd(c, f) = 1$ and $f^2 \equiv 1 \pmod c$,

$$T = \frac{V + c + Rf^2}{cf^2} > 0 \tag{18}$$

is an integer and realizes the required register V. These are the precise canonical guarantees used from package 61, including its signed index lemma, section 5.

Take $n = f + y - V - AI$. Its positivity requires proof. The canonical $R \ge 5$ and $s_0 = \Delta I \ge 2$ give

$$V \ge \frac{\chi_{s_0}(5)}{s_0} = 16s_0^4 - 20s_0^2 + 5 > s_0^4 = Q^2.$$

Also $f > (A-1)I$, so $AI - f < I$. From $d(y^2 - V^2) = V^2 - 1$, with $y > V \ge 2$ and $d = Q - 1 \ge 3$, we obtain $y < 2V$ and

$$y - V = \frac{V^2 - 1}{d(y+V)} > \frac{V}{4Q} > \frac{Q}{4} > I. \tag{19}$$

The last inequality uses $Q = \Delta^2 I^2$, $\Delta \ge 3$, $I \ge 9$. Therefore $n > 0$. Now $t = f + y - V$ and (17) give $U = Lf$, hence $N = 0$. This proof uses the actual positive gap; it needs no new ordinate-divisibility condition.

6.3 Ordered recovery from the adaptive carrier

Here are the essential soundness and completeness steps, with the order of divisibility recovery explicit. The full source-interface argument is retained in `positive12/PROOF.md` and its pinned source proofs.

First, $A_0 = 0$ makes $\tau = Z/M_0$ integral, by rational-square-root integrality. Reduction modulo q^3 and $q - 1$ gives

$$s = Y/q^3 \in \mathbb{Z}_{>0}, \qquad z_t = B_t/(q-1) \in \mathbb{Z}, \qquad \tau = \ell + k + r + s - Wz_t.$$

Since $\tau^2 = \ell(\ell + k) + 1 < (\ell + k)^2$, we have $Wz_t > r + s > 0$. Thus $B_t > 0$ and $C > 0$, before $q \mid X$ is known. The authentic mask bounds now give $3q < R < q^4 - 2$.

The compact radix has $\beta \equiv 1 \pmod 3$, so $J = 6j + 1$ gives $q \equiv 2 \pmod 3$. From $q^3 \mid Y$ and $H = 4Y(X + 1) + 3$,

$$\gcd(q, H) = 1, \qquad \gcd(S, H) \mid 3, \qquad \mu := \frac{S}{\gcd(S, H)} \geq \frac{4qY}{3} > R + 1.$$

Both S and M divide $4XY^2$ without assuming $q \mid X$. The equation $A_1 = 0$ makes $g = D_*/M$ integral. It is positive: $v < k$, $c > kY$ and $H < 6a$ give $D_* > ak(MY - 6) > 0$. We therefore have

$$g^2 - \Delta c^2 = 1, \qquad M \mid H(SX - v), \qquad \mu \mid v.$$

The first and main Pell equations give indices $n_0 < p$ such that $k = 2\psi_{2XY^2+1}(n_0)$ and $c = \psi_A(p)$. The case $n_0 = 1$ would force $\eta = \zeta = 1$ and $c = 2Y + 1 < \psi_A(2)$, so $n_0 \geq 2$. Hence $c > 8XY^3 \geq 8q^9 > 2R$, while Pell growth gives $c > 2p$. The normalized strong equation restored above supplies an index m with $c \mid m$ and $m > 2p$. The signed auxiliary-index lemma then recovers $p = R$ from $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. Every half-interval hypothesis has thus been established before the missing index quotient is restored.

The recurrence gives $k \equiv 2n_0 \pmod{4XY^2}$. Therefore $\mu \mid 2n_0 - R - 1$, and $n_0 < R$, $\mu > R + 1$ force $2n_0 = R + 1$. It follows that $v = k - R - 1$ is a positive multiple of $4XY^2$, hence of M. Only now does $M \mid H(SX - v)$ yield $q \mid HX$, and then $q \mid X$. The restored source coordinates are

$$w = X/q > 0, \qquad h = v/(XY) > 0, \qquad \gamma_{\mathrm{old}} = \gamma + w - v/M.$$

The last is integral and positive because $H\gamma_{\mathrm{old}} = g - ac - X > c - X > 0$. The first carrier restores the positive transport coordinate, and the normalized strong and auxiliary witnesses were restored in Section 6.2. The inherited compact source therefore yields a finite syntactic ZFC refutation.

Conversely, canonical source data have $q = B^N$ with odd $N > 1$, so their repunit J is 1 modulo six. They have $X = 2^R$, $Y = sq^3$, $w > q$ and v a positive multiple of $4XY^2$. With z_{old} the original transport coordinate, set

$$z_t = q(z_{\mathrm{old}} - 1) \geq q, \qquad r = Wz_t + \tau - \ell - k - s > qW - 2k > 0.$$

This gives $A_0 = 0$. Put $\gamma = \gamma_{\mathrm{old}} - w + v/M > 0$: indeed $v/M \geq XY/q^2 = wY/q \geq wq^2 > w$. Then $A_1 = 0$. Choose T, n by (18)–(19); these give $N = 0$. This proves the reverse existence implication. Negation gives Theorem 6.1.

6.4 Exact degree and arithmetic size

Let $q_0 = 6\beta j$, $k_0 = \eta + \zeta$, $R_4 = q_0^3(q_0 - F)$ and $B_{t,2} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F$. Exact cancellation yields the highest homogeneous parts

$$\begin{aligned}
[A_0]_{15} &= -2q_0^8 k_0^2 XY^2 B_{t,2}, \\
[A_1]_{13} &= 32q_0^2 k_0 X^2 Y^4 (4q_0^2 Y\gamma + R_4), \\
[N]_{82} &= 4i^{10} k_0^{20} X^{12} Y^{32} R_4^2.
\end{aligned}$$

For the last identity, the exact expansion in $\mathbb{Z}[A, i, c, b, n]$, with weights $(2, 1, 2, 4, 1)$, has 455 terms and leading form $4A^{12} i^{10} c^{20} b^2$. Substitution gives the displayed form. Their product is the degree-110 part of P_{12}; specifically,

$$[j^{18}\eta^{23} X^{16} Y^{39} i^{10} \alpha\gamma F^2] P_{12} = 1024(6\beta)^{18} > 0.$$

Thus the exact degree holds for the actual fixed coefficients. The delivered graph uses 57 multiplications, 29 additions and 16 subtractions. Its checker verifies exact identities, these leading forms, the nonzero coefficient, graph regeneration and all authentic coefficient bytes. The unbounded existence proof is the mathematical argument above; finite checks do not replace its inherited semantic dependencies.

7 An independent nine-variable improvement

The same snapshot permits a second result, obtained by composing three compatible previously separate constructions. Package 65 supplies the shared coefficient positions, package 63 supplies its deflated seventh-order root majorant, and package 62 supplies a shared addition chain for pure powers. The resulting single polynomial has nine strictly positive unknowns,

$$(b, g, h, s, w, C, i, j, n) \in \mathbb{Z}^9_{>0}, \qquad \deg F_9 = 1\,379\,730,$$

and a division-free evaluator with 478 binary operations: 286 multiplications, 163 additions, 29 subtractions. The nonnegative-coordinate translation has 485 operations and the same degree. This result improves package 67's lowest-degree nine-positive point, which has degree 1,494,006 and 487 operations. Lower-operation entries with higher degree remain useful.

7.1 A fully specified polynomial

The three source kernels D_0, D_1, D_2, 31 weights w_l and 22 targets q_l are the exact tables in `nine_positive/data/`. The four original31 coefficient ports are supplied as binary integers, with a finite reconstruction algorithm. Set $a = 2^{2134939}$ and $c = 2^{17079501}$. Define

$$\begin{aligned}
B &= ab^2, & \ell &= \sum_l B^{w_l}, & T_c^{\mathrm{mark}} &= \sum_l B^{q_l},\\
U_s &= \ell + g, & G_s &= (D_2(B)U_s + D_1(B))U_s + D_0(B), & N_0 &= B^{117},\\
M_s &= N_0 - 1 - (b-1)\ell, & S_s &= g + N_0(2G_s + T_c^{\mathrm{mark}}),\\
T_s &= M_s + N_0(B-2)T_c^{\mathrm{mark}}, & L &= bB^{1991}, & N &= cbB^{1057},\\
R &= L(S_s + T_s + 1) - S_s - 1, & t &= bB^{116}.
\end{aligned}$$

The following are polynomial abbreviations, not additional unknowns:

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M &= RY,\\
P &= 2M^2 U, & K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= bw, & J &= 4A - 5.
\end{aligned}$$

$$\begin{aligned}
Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2-1)C^2 + 1,\\
e &= iJC, & E &= eDC,\\
F &= (A^2-1)E^2 + 1, & L_m &= CD + 1 - A,\\
G &= A + FL_m, & H &= C + F(B_1 + (j-1)C),\\
I &= (G^2-1)H^2 + 1, & a_0 &= (P^2-1)K^2 + 1,\\
a_1 &= 4096^2 DFI.
\end{aligned}$$

Also put

$$\begin{aligned} m &= K^2(4t-g) - 16t(C-KY)^2, & V_0 &= PK, \\ \Omega &= Jm, & E_0 &= J(n+W_1^2)+Q, \\ N_* &= 2G + L_m, & N_1 &= 2e^2D(D-1)L_m + 3C, \\ N_2 &= 2e^2(D-1)L_m, & N_3 &= 2e^2(L_m - C), & N_4 &= 2e^2C. \end{aligned}$$

The majorant polynomial is

$$\begin{aligned} T = e\big[&(2048D^2 - 1024D - 256)N_* - 128N_1 - 80N_2 - 56N_3 - 42N_4 \\ &+ i^2(1568 - 448A + 288C)\big] + 640i + 4096. \end{aligned}$$

Finally set $V_1 = TH$, $\Theta = E_0 - \Omega(V_0 + V_1)$. The equation is

$$\boxed{F_9 = \big(\Theta^2 - \Omega^2(a_0+a_1)\big)^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{20}$$

All exponents are fixed and nonnegative. The literal arithmetic DAG, coefficient recipes and exact table entries remove any notational ambiguity.

7.2 Composition proof

The source 65 certificate proves 22 exact coefficient identities, including all absent monomials; 11,616 coefficient comparisons suffice. The 89 weighted records occupy 84 distinct positions. The weights have maximum 116 and the kernel degrees are $(1384, 1739, 1641)$, so the convolution height is 1873. Its unchanged positivity, weak-cap and signed-borrow estimates establish the exact source-to-binomial-divisibility interface required by the free-coordinate ratio finalizer.

The majorant 63 theorem is uniform in $A \geq 3$, $C, i \geq 1$:

$$T > 4096\sqrt{DF}\, G > 0.$$

It uses only the displayed Pell registers; it has no source-specific assumption on R, N, t. Its 264-term positive-coefficient certificate is included and independently re-expanded. Therefore $V_1 > \sqrt{a_1}$ and $V_0 > \sqrt{a_0}$ hold before decoding any zero.

At a zero of (20), $m \neq 0$, since $m = 0$ would give $F_9 = E_0^4 > 0$. Real factorization gives

$$E_0/J = m(V_0 + V_1 + u_0 + u_1), \qquad u_0^2 = a_0, \quad u_1^2 = a_1.$$

The majorants force $m > 0$. The inherited margin argument then recovers $C > K$ and $C > B_1 + W_1$, hence $a_1 > a_0$. The rational sum $u_0 + u_1 = \Theta/\Omega$ is nonzero; the identity $2(u_0+u_1)u_0 = (u_0+u_1)^2 + a_0 - a_1$ makes each root rational and therefore integral. The same equation gives $J \mid Q$. These are exactly the ratio conditions that recover a binary source witness. Conversely, a source zero supplies the cofinal ratio witnesses, and

$$n = m\big(V_0 + V_1 + \sqrt{a_0} + \sqrt{a_1}\big) - W_1^2 - Q/J$$

is integral and strictly positive. It produces a zero of (20). The unchanged native compiler and syntactic-refutation calibration then yield the required consistency equivalence in ordinary mathematics formalizable in ZFC. No new end-to-end proof-assistant derivation is claimed.

7.3 Exact degree and arithmetic cost

Write $r = \deg R = 7963$, $q = \deg N = 2115$, $W = 116$. Then

$$\deg A = r + 4q + 2 = 16425, \quad \deg H = 139373, \quad \deg \Omega = 82360.$$

The exact majorant remainder has leading monomial $2880A^{10}C^{12}i^3$. Its weighted degree is 164265. The corresponding defect identities give

$$\deg(T^2 - 4096^2 DFG^2) = 246399, \qquad \deg(V_1^2 - a_1) = 525145.$$

For $\delta_k = V_k^2 - a_k$, put $B_* = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1)$. The exact identity

$$F_9 = B_*^2 + 4\Omega^2 V_0 V_1 B_* + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1)$$

resolves the apparent leading cancellation in the primary norm. Its first term alone has highest degree, so

$$\deg F_9 = 86r + 328q + 8W + 264 = 1\,379\,730.$$

The diagonal leading coefficient is $2025 \cdot 2^{188} a^{689526} c^{328} > 0$, certifying that this upper bound is attained for the actual fixed coefficients. A separate degree-checking DAG computes $D^{20}F_9$; its larger size is not charged as the primary evaluator.

The initial composition has 489 operations. The checked global power chain reduces this to 478. Each rewrite is certified by an integer equality $e = e_1 + e_2$ and an exact structural comparison of the remaining DAG. Forty modular evaluations provide supplementary transcription diagnostics; the source, majorant, degree and rewrite identities are checked exactly.

8 Proof status, alternatives, and integration

8.1 The precise proof boundary

The new determinant lemma, its canonical coefficient bounds, and the integer elimination maps are elementary unbounded arguments. The two positive companions also include complete ordinary arguments, including the order in which denominator, sign, rank and divisibility facts are recovered. All use finite integer arithmetic, elementary real inequalities or standard number-theoretic lemmas that can be formalized in ZFC.

The inherited statements are explicitly the following: the finite native encoding is sound for blank K215 accepting computations; every finite halt has the canonical padded and rank-tuned native witnesses; the cited Pell and ratio interfaces preserve existence; and the calibrated machine has a finite blank halt exactly when the fixed ZFC calculus has a finite refutation. The new code does not prove these facts by running a bounded machine search.

The active repository distinguishes a closed generic nine-variable Lean equation from the optimized native circuits, for which the authentic native calibration in Lean and a derivation in the object-language ZFC calculus remain open [8]. Here the native bridge is imported from the cited packages as an ordinary mathematical proof. This archive supplies *no newly compiled end-to-end Lean or Rocq certificate*, and no exported derivation in the repository's object-level ZFC syntax. Ordinary mathematical composition and an object-language proof artifact are different deliverables. The statement in Theorem 1.1 has the former evidence, together with exact finite certificates for its explicit data.

Negating the proved existence equivalence gives the requested statement. It does not prove consistency. If ZFC is consistent, the new equations have no zeros in their stated domains; that conditional assertion does not entail a separate proof of their unsolvability inside ZFC.

8.2 What the finite checks establish

The root verification command rebuilds the coefficient tuple and polynomial packets and checks the following independent evidence:

- All 430 literal machine transitions reconstruct the exact 943-tile list, and all four old coefficient bytes match their pinned hashes.
- The new scales, leading support, unique top convolution pair, and strict carry and balancing hypotheses hold for those actual integers.
- The quartic, sextic and octic have exactly the stated live unknowns, residual counts, exact degrees and nonzero leading coefficients.
- Every monic elimination and every complete inverse restoration is an exact symbolic integer-polynomial identity.
- A separately written checker, without importing the primary sparse engine, repeats the integer identities, coefficients and literal formulas.
- The twelve-positive circuit has the exact degree-110 certificate and 102 live operations. The nine-positive component checks all 11,616 source coefficient comparisons, the 264-term majorant certificate, degree-defect identities and an exact shared-power rewrite.

Finite sign boxes and modular evaluations are labeled as transcription diagnostics. They are not substituted for universal proofs. The checks also run with Python's `-O` flag; acceptance does not rely on removable assertions.

8.3 Why direct proof encoding was not promoted

The request allowed a route independent of K215 and Jones. That direction was investigated. Carl and Moroz give an explicit Diophantine representation of predicate-calculus provability and specialize it to finitely axiomatized Gödel–Bernays set theory [9]. Their stated construction uses 3,639,528 positive witnesses before a four-square translation. It is a concrete alternative endpoint, but is not competitive with this repository's present size bounds. Merely replacing the words "Turing machine" by "proof checker" would not supply a smaller polynomial.

The determinant reduction itself does not use Jones's universal polynomial or Sun's variable reduction; its low-degree forms inherit the native Pell source. The nine-positive companion retains the existing Jones-style ratio reduction. Sun's 2026 ten-integer result [10] remains relevant to the separate minimum-integer-variable frontier. No new eight-positive or nine-unrestricted-integer construction was obtained.

8.4 A useful next mathematical target

The structural lesson is specific: when canonical encoded quantities have fixed leading digits, one should first look for a low-degree definite-form region containing all those canonical values. Several independent positivity witnesses may then be replaced by one. Arbitrary source zeros need only map soundly; completeness can regenerate canonical witnesses. The present proof works because two input words have fixed leading digit positions and coefficients, unaffected by all padding and rank control.

Extending that mechanism to other guards requires a new uniform canonical range theorem. The ratio $\eta/(\eta+\zeta)$ in the Pell part, for example, has not been given an adequate fixed-coefficient

balancing theorem here. Dropping its remaining product guard would be unjustified. Similarly, a different native tuple needs a fresh leading-support and carry audit before using these two scales. This is a concrete research direction, not an additional claimed result.

8.5 Repository use and reproduction

The archive is an independent research package. It need not replace the active formally checked equation. Its root commands are

```
python3 verify_all.py
python3 -O verify_all.py
python3 build_report.py
```

The mathematical replay uses Python's standard library. A TeX Live installation is needed only to recompile the article. Each positive companion keeps its own complete coefficient binding: its ports must not be mixed with the main integer tuple or with each other. The repository snapshot, source identities, licenses, notices, deterministic polynomial packets and verification receipts are included. The next numbered package directory can be chosen at incorporation time.

A Finite definition of the main native coefficients

This appendix, together with the literal 943-tile JSON file, specifies every coefficient of the integer equations by bounded finite arithmetic. The original builder and independent raw hashes are supplied unchanged. Let the ordered tiles be $(x_{it})_{0\le i<943,\ 0\le t<4}$, with Start at index zero and Forbidden at index one. Put

$$e_t(j) = o_t + 2j, \quad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960), \qquad W_j = 2^{10+2j}.$$

The support is

$$\mathcal{E} = \{0, \dots, 942\} \cup \{e_t(j) : t < 4, j < 10\} \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.$$

For each tile set

$$c_i = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) W_{10t+j} + \sum_{j=40}^{43} W_j.$$

Set $c_{e_t(j)} = W_{10t+j}$, set the coefficients at 1095,1096,2195,3296 to $W_{40}, W_{41}, W_{42}, W_{43}$ respectively, and set $c_{1063} = c_{1064} = 0$. Also put

$$\mu = 1022 + \sum_{j=0}^{43} W_j, \quad Z_0 = (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}.$$

Then

$$D_C = \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),$$

$$D_R = \rho,$$

$$M_C = B - 1 - \sum_{\substack{e\in\mathcal{E} \\ e\neq 1}} \rho^e - 2\rho^{1063},$$

$$M_{F,n} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) + \rho^{2195} + \rho^{3296} + Z_0.$$

The four inherited ports are exactly

$$\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_{F,n} + B - 1.$$

In particular M_F is *already shifted*. The two extra ports are (4). The literals contain no undecided search or parameter that depends on whether ZFC is consistent. The compiler's finite geometry, parity and rank-control proof is retained separately; a tuple with the same rough numerical bounds would not be an adequate substitute.

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